

Lecture 2 $\Rightarrow$ Modulation.

Analog modulation $\begin{cases} \rightarrow \text{AM} \rightarrow \text{amplitude Modulation} \\ \rightarrow \text{FM} \rightarrow \text{Frequency} \\ \rightarrow \text{PM} \rightarrow \text{phase} \end{cases}$

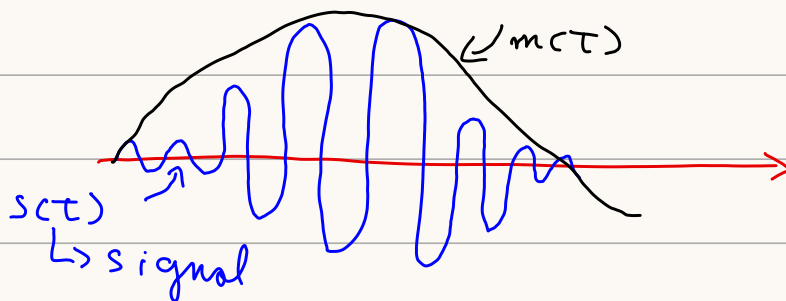
Simple Analog modulation is achieved by modulating a carrier by The message signal

$$s(t) = m(t) \cdot A_c \cdot \cos[2\pi f_c t]$$

$e^{2\pi i f_c t} \Rightarrow \delta[F - F_c]$
impulse

$$\rightarrow \frac{A_c}{2} [e^{2\pi i f_c t} + e^{-2\pi i f_c t}]$$
$$\rightarrow S(F) = \frac{A_c}{2} [M[F + F_c] + M[F - F_c]]$$

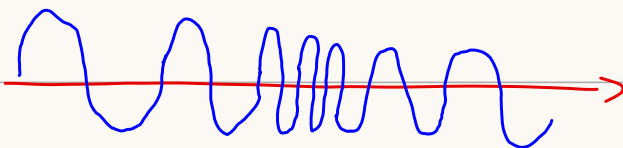
Example of AM signal



FM example:

$$\frac{d\phi}{dt} = \text{Freq}$$

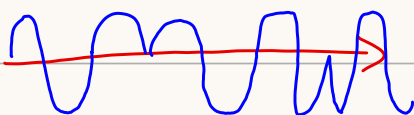
$$s(t) = A \cos [2\pi f_c t + 2\pi K_f \int_{-\infty}^t m(\tau) d\tau]$$



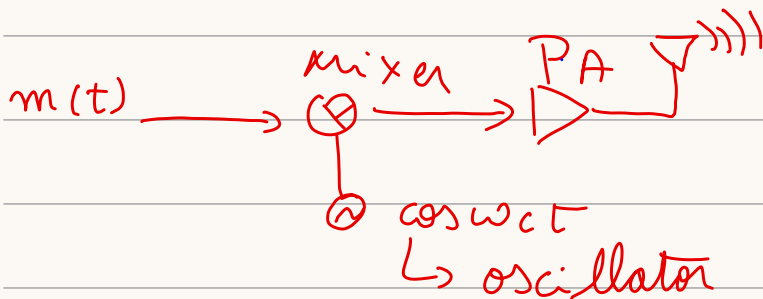
info in Frequency #

PM example:

$$A \cos [\omega_c t + 2\pi K_m m(t)]$$



up conversion modulation:



in up conversion we use
PLL $\Rightarrow$ phase locked
Loop.

Demodulation:-

$$s(t) = A_1 m(t) \cos(\omega_c t)$$

$$v(t) = s(t) \times \cos \omega_c t \times A_2$$

$$= \frac{A_1 A_2}{2} [m(t) + \boxed{m(t) \cos[2\omega_c t]}] \rightarrow \text{Image problem}$$

$\hookrightarrow$ Lpf to get needed signal

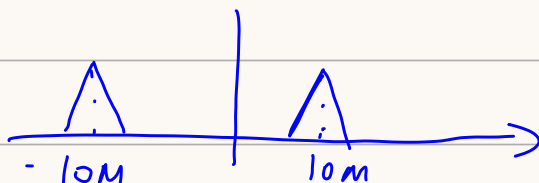
Image problem:

assume $s(t) = A_1 \cos[1010 \text{ MHz}] \times 2\pi \Rightarrow f_{LO} = 1000 \text{ MHz}$

Downcon $v(t) = A_1 \cos[1010 \text{ MHz}] \times A_2 \cos[1000]$

$$= \frac{A_1 A_2}{2} [\cos[2010 \text{ MHz}] + \cos[10 \text{ MHz}]]$$

$\downarrow$



$$\delta[f - 10 \text{ MHz}] + \delta[f + 10 \text{ MHz}]$$

IT will create interference @ 10 MHz ?? $\leftarrow 990 \text{ MHz}$

$\hookrightarrow$ If there was a signal @ 990 MHz

